

Applying Machine Learning to Diagnose Thyroid Cancer Nodules

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Problem Overview:

Each year, over 40,000 cases of thyroid cancer are diagnosed in the United States [1]. Thyroid cancer is a disease characterized by the uncontrolled growth of cells originating in the thyroid gland. Most thyroid cancers do not produce noticeable signs or symptoms in the early stages of the disease. Later on in disease progression, symptoms of thyroid cancer include an individual having a noticeable lump in the neck, changes to one's voice, and trouble swallowing [2].

Most cases are treated with surgery and have a 5-year relative survival of 98.5% [1]. Recent advances in artificial intelligence have attempted to apply machine learning to predict whether an individual's thyroid nodule is benign or malignant based on patterns in recorded health data (e.g., free triiodothyronine (T3) level, gender) in patient data [3]. Machine learning models can be trained on historical patient thyroid data to classify a given patient's thyroid nodule as benign or malignant, providing an opportunity to improve diagnostic precision and enhance clinical decision-making [4]. Overall, machine learning has the potential to reduce unnecessary biopsies [5], unnecessary surgeries [6], and improve personalized treatments for thyroid cancer patients [7]. This motivates our research question:

How well can machine learning models classify thyroid nodules as benign or malignant using patient data?

The goal of our analysis is to use a Kaggle dataset that includes patient features (e.g. Age, Gender, Country, and thyroid stimulating hormone (TSH) Level) that span real-world risk factors for thyroid cancer along with the patient's corresponding thyroid nodule diagnosis [8], and apply machine learning models to the dataset to try to predict each patient's thyroid nodule diagnosis based on the predictor variables. Our goal is to match our predictions as closely as possible to the actual values in the "Diagnosis" column of this dataset.

Description of the Dataset:

The dataset is simulated from real-world thyroid cancer risk factors and comprises 212,691 rows (212,691 unique patients) and 17 columns (recorded patient features). This dataset specifically examines certain ethnic groups, nationalities, and individuals with risk factors such as radiation exposure, iodine deficiency, and a family history of the condition. The columns include: Patient ID, Age, Gender, Country, Ethnicity, Family History of Thyroid Cancer (binary yes/no), Previous Radiation Exposure (binary yes/no), Iodine Deficiency (binary yes/no), Smoking (binary yes/no),

Obesity (binary yes/no), Diabetes (binary yes/no), TSH Level, Free T3 Level, Total Thyroxine (T4) Level, Nodule Size, Thyroid Cancer Risk (low, medium and high categories) and Diagnosis (benign and malignant categories).

TSH Level, Free T3 Level, and Total T4 Level are columns that correspond to measures of the amount of hormones produced by a given patient's thyroid gland [9]. TSH Level measures a patient's level of a hormone released by the pituitary gland [10]. Free T3 Level measures a patient's T3 level, a hormone produced by the thyroid [11]. Total T4 Level measures a patient's level of Thyroxine, another hormone produced by the thyroid [12].

Below describes the data types of the initial loaded dataframe:

Patient_ID	object
Age	int64
Gender	object
Country	object
Ethnicity	object
Family_History	object
Radiation_Exposure	object
Iodine_Deficiency	object
Smoking	object
Obesity	object
Diabetes	object
TSH_Level	float64
T3_Level	float64
T4_Level	float64
Nodule_Size	float64
Thyroid_Cancer_Risk	object
Diagnosis	object

Feature engineering was applied to extract additional variables of “Risk Factor Score”, “TSH Level and Thyroid Nodule Size Interaction”, and “Free T3 to TSH Level Ratio”. Variables of Patient ID and Thyroid Cancer Risk were dropped from the data.

The final set of input features were Age; Gender (one-hot encoded); Country (one-hot encoded); Ethnicity (one-hot encoded); Family History of Thyroid Cancer (one-hot encoded); Previous Radiation Exposure (one-hot encoded); Iodine Deficiency (one-hot encoded); Smoking (one-hot encoded); Obesity (one-hot encoded); Diabetes (one-hot encoded); TSH Level; Free T3 Level; Total T4 Level; Nodule Size; Risk Factor Score; TSH Level and Thyroid Nodule Size Interaction; and Free T3 to TSH Level Ratio.

How to run the code:

The code is a series of Jupyter notebooks. Two of the notebooks correspond to Exploratory Data Analysis and to preparing the data for machine learning. There also exists one notebook for each model. Depending on the model, different packages must be loaded.

Workflow:

- 1) run “pip install -r requirements.txt” to make sure installed packages on computer meet the required packages required to run our code
- 2) data_cleaning_exploration/data_preparation.ipynb
- 3) data_cleaning_exploration/exploratory_analysis.ipynb

The following is up to the user's interests:

- Random Forest: audrey_model.ipynb
- XGBoost: chrissie_model.ipynb
- Logistic Regression: emma_model.ipynb
- Neural Networks: tara_model.ipynb, audrey_regular_nn_model.ipynb

Packages Include:

Numpy

Random

Pandas

Sci-kit Learn

Tensorflow

Matplotlib

Xgboost

Pathlib

Joblib

Jupyter

Model Versions

Numpy (v.1.24.0)

Pandas(v.2.0.0)

Matplotlib(v.3.7.0)

Scikit-learn(v.1.3.0)

Tensorflow(v.2.15.0)

Jupyter(v.1.15.0)

Decisions:

The outcome we wanted to predict was Diagnosis. No missing values were present in the data, so missing values did not need to be addressed. Patient ID and Thyroid Cancer Risk were excluded from our predictors. Patient ID was removed because it is a variable that assigns a unique identifier to each patient, and does not offer useful information to help us decide what a patient's thyroid nodule diagnosis is. Thyroid Cancer Risk was removed because it would create a data leakage and potential for circulatory analysis, as Thyroid Cancer Risk describes very similar information to a patient's thyroid nodule diagnosis (patients with high thyroid cancer risk are more likely to have their thyroid nodule be diagnosed as malignant).

Feature engineering was applied to existing columns in our data in order to transform raw data into more meaningful insights that can better inform machine learning models that are applied to the data. Three additional features were created through feature engineering: Risk Factor Score; TSH Level and Thyroid Nodule Size Interaction; and Free T3 to TSH Level Ratio. “Risk Factor Score” was created by assigning a value of 1 for each of the following six thyroid cancer risk factors—Family History of Thyroid Cancer, Previous Radiation Exposure, Iodine Deficiency, Smoking, Obesity, and Diabetes—when the patient’s value for the variable is “Yes”, and 0 when the value is “No”, then summing the values across these variables for each patient to calculate their overall Risk Factor Score. The variable Risk Factor Score was added in order to summarize six binary risk factors for thyroid cancer into a more informative single score. “TSH Level and Thyroid Nodule Size Interaction” was created by multiplying each patient’s value for TSH Level and Thyroid Nodule Size together. This interaction term was included to capture the potential dependency between these two variables, as the effect of one may vary depending on the value of the other. “Free T3 to TSH Level Ratio” was created by dividing each patient’s Free T3 value by their TSH Level value. This feature was added to better capture the ways in which a patient’s values for these variables may be physiologically linked. Standard Scaler was applied to the numerical features to standardize numerical features to have a mean of 0 and a standard deviation of 1. One-hot encoding was applied to the categorical features to convert them into numerical binary formats interpretable by our machine learning models.

We used the following set of predictors when training machine learning models in order to predict the outcome variable: Age; Gender (one-hot encoded); Country (one-hot encoded); Ethnicity (one-hot encoded); Family History (one-hot encoded); Previous Radiation Exposure (one-hot encoded); Iodine Deficiency (one-hot encoded); Smoking (one-hot encoded); Obesity (one-hot encoded); Diabetes (one-hot encoded); TSH Level; Free T3 Level; Total T4 Level; Nodule Size; Risk Factor Score; TSH Level and Thyroid Nodule Size Interaction; and Free T3 to TSH Level Ratio. We decided to not perform feature selection, as we thought that the aforementioned predictors were convincing features to train our models on because they correspond to well-known thyroid cancer risk factors.

We decided to work with 5 different categories of machine learning models to classify thyroid nodules as being benign or malignant based on patient data: Logistic Regression, Random Forest, XGBoost, regular Neural Network, and Feedforward Neural Network (FFNN). Logistic Regression was employed as a baseline model for binary classification. Random Forest was utilized because it is an ensemble learning method that may be more robust to overfitting and better equipped to capture complex, non-linear interactions between features. XGBoost was selected because it is another ensemble learning method that may be robust to overfitting and can potentially reduce bias through its gradient boosting algorithm. Regular Neural Network was chosen because of its ability to capture complex, non-linear relationships in high-dimensional

data. FFNN was utilized because of its ability to learn relevant feature representations from the input features, potentially identifying additional valuable patterns in the data.

The dataset was randomly partitioned into training and test sets, with 70% allocated for training and 30% for testing. Stratified 5-fold cross validation was applied on the training data to evaluate model performance across multiple subsets of the training data in order to reduce the risk of overfitting, and to ensure that the proportion of benign and malignant cases within each fold mirrors that of the overall training dataset. During the model training process conducted on the training data, hyperparameter tuning was employed to systematically search for the combination of hyperparameter values that yielded the best model performance. Error metrics of accuracy, precision, recall, F1 score, and AUC were used to compare the performance of our models. We decided to use the error metric accuracy to better understand how many correct predictions our models make. We employed precision to inspect more in-depth the false positives made by our models, recall to evaluate more closely the false negatives made by our models, and F1 score to evaluate how well our models balance precision and recall. AUC was selected as an error metric to assess how well our models distinguish between classes. For each model, error metrics were calculated on the training set by averaging per-fold error metrics across all folds of the training data. During hyperparameter tuning, for each of the five categories of models, we selected the model with the highest average 5-fold cross validated AUC on the training set to apply to the test set. After, we calculated error metrics for each “best model” for each model category on the test set. We then selected the model that achieved the highest AUC on the test set as our final top model for predicting whether a thyroid nodule is benign or malignant.

It is important to note that we did not have the time to better optimize our models so that they produce stronger predictions. A future step is to spend more time optimizing our models, as it is crucial that our models avoid misclassification as much as possible given the context of classifying patient thyroid nodules. In addition, we were interested in conducting further investigation into the factors driving the models’ performance on the thyroid cancer dataset; however, we decided to not add this additional analysis due to time constraints.

Example Output:

Logistic Regression Model Outputs:

Figure 1. ROC Curve for Best Logistic Regression Model

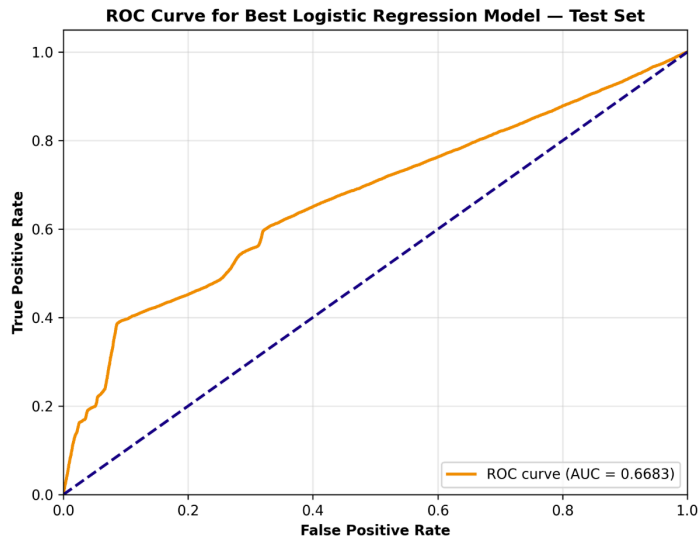


Table 1. Model Metrics for Logistic Regression Model

Accuracy	0.7803
Precision	0.6905
Recall	0.1004
F1	0.1753
AUC	0.6683

Random Forest Model Outputs:

Figure 2. ROC Curve for Best Random Forest Model

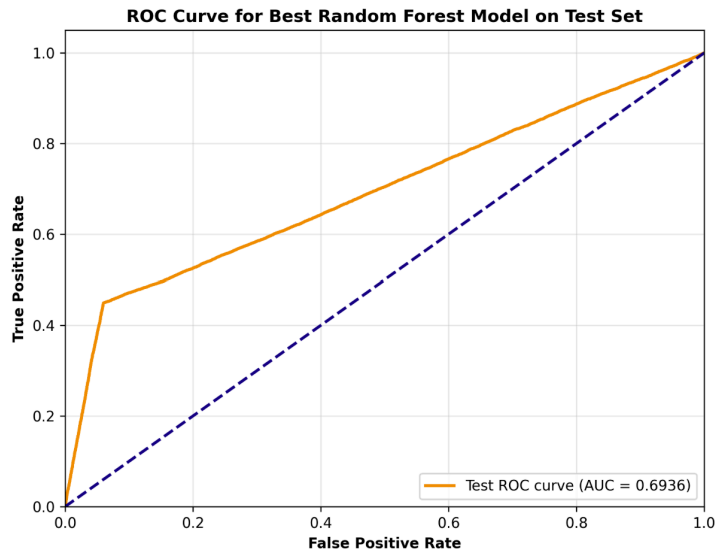


Table 2. Model Metrics for Random Forest Model

Accuracy	0.8256
Precision	0.6945
Recall	0.4469
F1	0.5439
AUC	0.6936

XGBoost Model Outputs:

Figure 3. ROC Curve for Best XGBoost Model

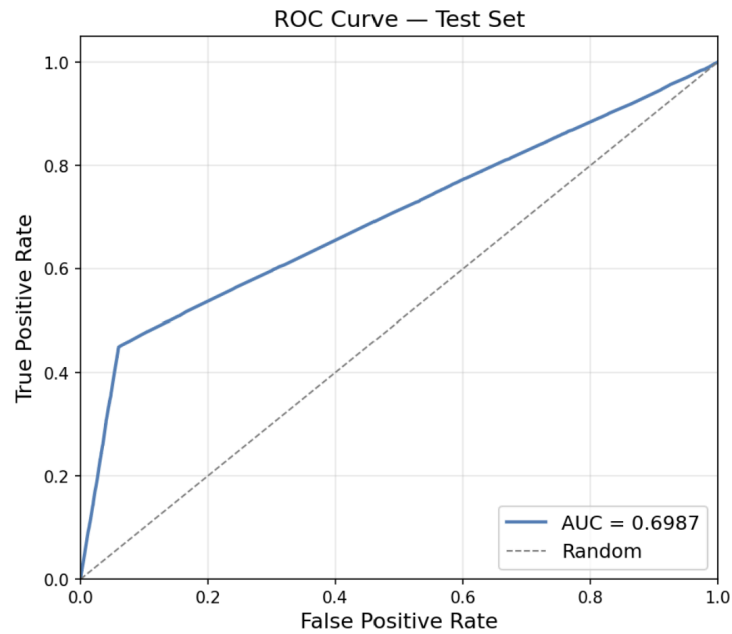


Table 3. Model Metrics for XGBoost Model

Accuracy	0.8238
Precision	0.6949
Recall	0.4322
F1	0.5330
AUC	0.6987

Neural Networks Model Outputs:

Regular Neural Network

Figure 4. ROC Curve for Best Regular Neural Network Model

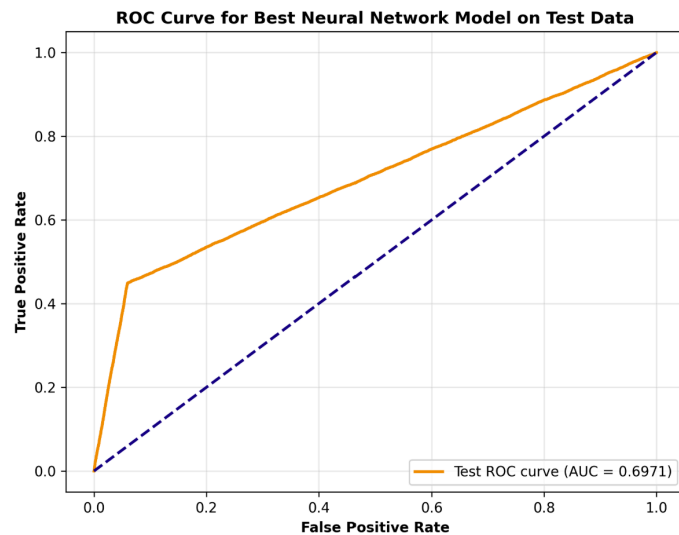


Table 4. Model Metrics for Regular Neural Network Model

Accuracy	0.8259
Precision	0.6952
Recall	0.4478
F1	0.5447
AUC	0.6971

Feed-Forward Neural Network

Figure 5. ROC Curve for Best Feed-Forward Neural Network Model

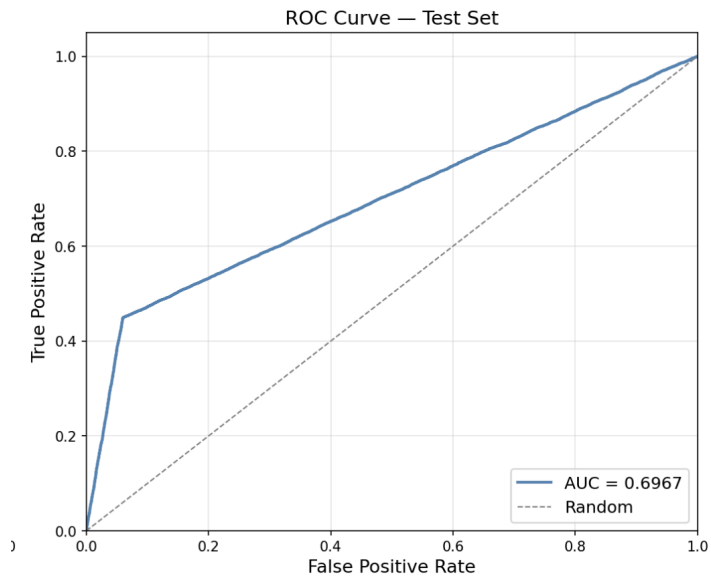


Figure 5. Model Metrics for Feed- Forward Neural Network Model

Accuracy	0.8259
Precision	0.6946
Recall	0.4485
F1	0.5451
AUC	0.6967

Table 6. Final Table of All Model Metrics

	Logistic Regression	Random Forest	Neural Network (Feed Forward)	Neural Network	XGBoost
Accuracy	0.7803	0.8256	0.8259	0.8259	0.8238
Precision	0.6905	0.6945	0.6946	0.6952	0.6949
Recall	0.1004	0.4469	0.4485	0.4478	0.4322
F1	0.1753	0.5439	0.5451	0.5447	0.5330
AUC	0.6683	0.6936	0.6967	0.6971	0.6987

Discussion:

Based on the 5 models' metrics, we selected XGBoost as the top model because it had the highest AUC (0.6987). We note that the AUCs for the 5 models are fairly similar (range: 0.6683-0.6987). A future avenue is to explore more about why this occurred.

One benefit of using an XGBoost model is that it is widely-known to perform well on structured, tabular data, which can be valuable, as we worked with such data in our analysis. XGBoost is also highly resistant to overfitting due to its built-in regularization parameters. It can also capture complex, non-linear relationships between features that simpler models, such as logistic regression, may miss. Finally, XGBoost is highly computationally efficient, which is crucial for analyzing large datasets like ours.

Although XGBoost modeling has many strong features, it also has significant limitations. It can be difficult to explain the results of our analyses to clients and patients because of their limited interpretability. Also, the highest AUC was only 0.6987, indicating that there is definitely room for improvement and that the model may need to be revised to be more reliable. The model also demonstrated low precision, recall, and F1 scores, indicating that it often fails to correctly identify true positive malignant cases. This is very concerning in the medical context and can have very serious consequences for patient outcomes. Finally, class imbalance is another limitation, as XGBoost may struggle to correctly classify underrepresented classes when the dataset is disproportionately skewed toward benign or malignant cases.

Citations:

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